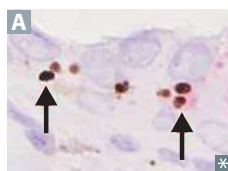


► PATHOLOGY—AGING

Normal aging	Time-dependent progressive decline in organ function resulting in ↑ susceptibility to disease. Associated with genetic (eg, telomere shortening), epigenetic (eg, DNA methylation), and metabolic (eg, mitochondrial dysfunction) alterations.
Cardiovascular	↓ arterial compliance (↑ stiffness), ↑ aortic diameter, ↓ left ventricular cavity size and sigmoid-shaped interventricular septum (due to myocardial hypertrophy), ↑ left atrial cavity size, aortic and mitral valve calcification, ↓ maximum heart rate.
Gastrointestinal	↓ LES tone, ↓ gastric mucosal protection, ↓ colonic motility.
Hematopoietic	↓ bone marrow mass, ↑ bone marrow fat; less vigorous response to stressors (eg, blood loss).
Immune	Predominant effect on adaptive immunity: ↓ naïve B cells and T cells, preserved memory B cells and T cells. Immunosenescence impairs response to new antigens (eg, pathogens, vaccines).
Musculoskeletal	↓ skeletal muscle mass (sarcopenia), ↓ bone mass (osteopenia), joint cartilage thinning.
Nervous	↓ brain volume (neuronal loss), ↓ cerebral blood flow; function is preserved despite mild cognitive decline.
Special senses	Impaired accommodation (presbyopia), ↓ hearing (presbycusis), ↓ smell and taste.
Skin	Atrophy with flattening of dermal-epidermal junction; ↓ dermal collagen and ↓ elastin (wrinkles, senile purpura), ↓ sweat glands (heat stroke), ↓ sebaceous glands (xerosis cutis). ▪ Intrinsic aging (chronological aging)—↓ biosynthetic capacity of dermal fibroblasts. ▪ Extrinsic aging (photoaging)—degradation of dermal collagen and elastin from sun exposure (UVA); degradation products accumulate in dermis (solar elastosis).
Renal	↓ GFR (↓ nephrons), ↓ RBF, ↓ hormonal function. Voiding dysfunction (eg, urinary incontinence).
Reproductive	Males—testicular atrophy (↓ spermatogenesis), prostate enlargement, slower erection/ejaculation, longer refractory period. Less pronounced ↓ in libido as compared to females. Females—vulvovaginal atrophy; vaginal shortening, thinning, dryness, ↑ pH. Due to ↓ estrogen from exhaustion of ovarian follicles (menopause).
Respiratory	↑ lung compliance (↓ elastic recoil), ↓ chest wall compliance (↑ stiffness), ↓ respiratory muscle strength; ↓ FEV ₁ , ↓ FVC, ↑ RV (TLC is unchanged); ↑ A-a gradient, ↑ $\dot{V}/\dot{Q}$ mismatch. Ventilatory response to hypoxia/hypercapnia is blunted. Less vigorous cough, slower mucociliary clearance.

Lipofuscin

A yellow-brown, “wear and tear” pigment **A** associated with normal aging.

Composed of polymers of lipids and phospholipids complexed with protein. May be derived through lipid peroxidation of polyunsaturated lipids of subcellular membranes.

Autopsy of older adult will reveal deposits in heart, colon, liver, kidney, eye, and other organs.